
Development and Local Knowledge

New approaches to issues in
natural resources management,
conservation and agriculture

Edited by Alan Bicker,
Paul Sillitoe and Johan Pottier

 **Routledge**
Taylor & Francis Group
LONDON AND NEW YORK

First published 2004

by Routledge

11 New Fetter Lane, London EC4P 4EE

Simultaneously published in the USA and Canada

by Routledge

29 West 35th Street, New York, NY 10001

Routledge is an imprint of the Taylor & Francis Group

© 2004 Alan Bicker, Paul Sillitoe and Johan Pottier for editorial matter; articles © original contributors.

Typeset in Galliard by Wearset Ltd, Boldon, Tyne and Wear

Printed and bound in Great Britain by Antony Rowe Ltd, Chippenham, Wiltshire

All rights reserved. No part of this book may be reprinted or reproduced or utilized in any form or by any electronic, mechanical, or other means, now known or hereafter invented, including photocopying and recording, or in any information storage or retrieval system, without permission in writing from the publishers.

British Library Cataloguing in Publication Data

A catalogue record for this book is available from the British Library

Library of Congress Cataloging in Publication Data

A catalog record for this book has been requested

ISBN 0-415-31826-2

The knowledge of indigenous desire

Disintegrating conservation and development in Papua New Guinea

Colin Filer

Talk of indigenous knowledge

How can Western conservationists talk to Melanesian landowners about 'indigenous knowledge', when Melanesian landowners do not think of themselves as 'indigenous people', and would rather talk about that Western form of knowledge that commands the gateway to 'development'? How could Jared Diamond (1997) answer Yali's question in a way that would make Yali think again about the value of the knowledge that had failed to yield the 'cargo'? Such is the question which hangs over the battle for 'biodiversity values' in the lowland rainforests of Papua New Guinea (PNG). At the level of policy, this is a battle between a group of Occidental donors, led by the World Bank, and a group of Oriental loggers, led by Rimbunan Hijau. The choices made by Papua New Guineans, both at the level of the state and at the level of the village, have become the prizes in this tug of war.

Over the past decade, the PNG government has been persuaded to impose severe restrictions on the further expansion of the logging industry, but these policy measures have not stopped the loggers from making promises of development to local landowners whose forests have not yet been logged, and they have not enhanced the government's own capacity to persuade the landowners to stop listening to such promises. So this latter task has been left to Western conservationists, whose projects are thus designed to win back the hearts and minds of landowners who tend to blame their own government for their own lack of development. The landowners count as landowners because the point at issue here is the use of customary land, on which more than 99 per cent of the country's natural forests happen to grow. Indeed, they have come to think of themselves as landowners, or sometimes as resource owners, because they have come to believe that their best chance of development is to sell the natural resources that their land contains (Filer 1997). But their determination to defend their customary title to the land itself has also grown apace, because they do not trust the government to do so, and because its loss would leave them powerless and impoverished (Ballard 1997). Their land is their last card in the gamble for modernity, and

their status as landowners is critical to their self-esteem, if not to their standard of living.

The broad political setting of the conservation business in PNG can thus be represented as a dialogue between four parties, where the voice of the PNG government is aligned with the global interest of the donor community, but where landowners (and their political leaders) have the upper hand in deciding the voice to which they will listen, and where conservationists must find a path between the empty promises of both the government and the developers to get their message across to the custodians of biodiversity (Figure 5.1). In this context, it is understandable that conservationists have been attracted to the global fashion for 'integrated conservation and development projects' (ICDPs), recognizing that they may well have to make some promise of development to local communities in order to compensate for the government's inability to enforce the conservation option. On the other hand, the prevalence of customary tenure makes it difficult or impossible to establish buffer zones around protected areas, of the kind which have been advocated by proponents of ICDPs in other parts of the world. In other words, the trade-off between conservation and development cannot simply be construed as a sort of land use planning exercise, with or without local participation, but has to be adapted to the social realities of the Melanesian landscape.

In this context, we might say that local landowners or resource owners also count as indigenous people, even if they do not use a comparable phrase themselves, because such people are normally defined, at least in part, by their 'close attachment to ancestral territories and to the natural resources in these areas' (World Bank 1991). But most definitions of indigenous peoples also place some emphasis on their subordinate political status, as ethnic or tribal minorities which are distinct from the dominant society, and the concept of a 'dominant society' does not help us to understand the role of landowners, or for that matter the role of national politicians, bureaucrats and conservationists – who are also landowners – as arbiters in a tussle between Occidental donors and

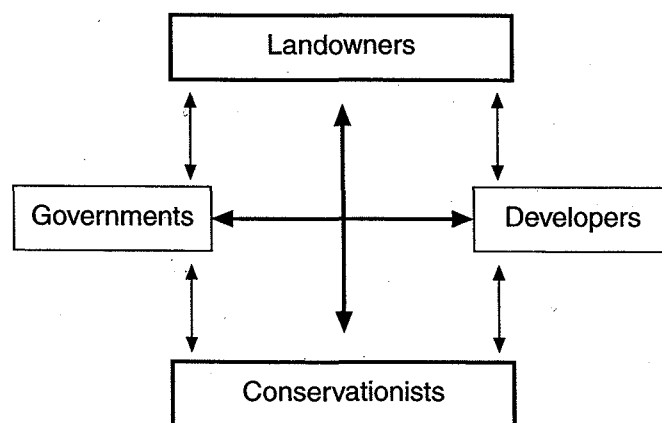


Figure 5.1 The central axis of stakeholder relationships in the PNG conservation business.

Oriental loggers. Furthermore, the idea of a 'close attachment to ancestral territories' can all too easily beg one of the main questions which I wish to pose in my account of this struggle, by leading us to assume that these indigenous people are also 'rainforest people', with a traditional interest in the conservation of natural forests or forest species.

This is where the subject of 'indigenous knowledge' enters the picture, as the topic of a dialogue within a dialogue, a type of scene or sub-plot in the public or private performance of conservation policy. It might be argued that the value and scope of indigenous knowledge has already been diminished or distorted once local actors assume the character of landowners or villagers, rather than indigenous people. But we do not have to assume that all indigenous knowledge belongs to indigenous people, that indigenous knowledge is necessarily distinct from other forms of knowledge, or that any form of knowledge is the special property of an equivalent class of people. The semantic point at issue is connected to the economics and politics of conservation in a specific regional and sectoral policy domain.

A lot of talk about the value of indigenous knowledge takes place when the pronouncements of government policy and the blandishments of logging companies are both out of earshot, on the other side of the battlefield, and members of the conservation community are divided by the roles that they play in the act of communication with local landowners. On one side are those Melanesian conservationists who are employed as the frontline foot soldiers of donor-funded conservation projects. On the other side are those Western anthropologists who may appear as listeners or participants in the frontline conversation, as passive scientists or active consultants, and sometimes even as project managers or technical advisers. And behind them are those Western conservationists who are not anthropologists, and whose ultimate aim is to address the needs of nature rather than the aspirations of its local guardians (Figure 5.2).

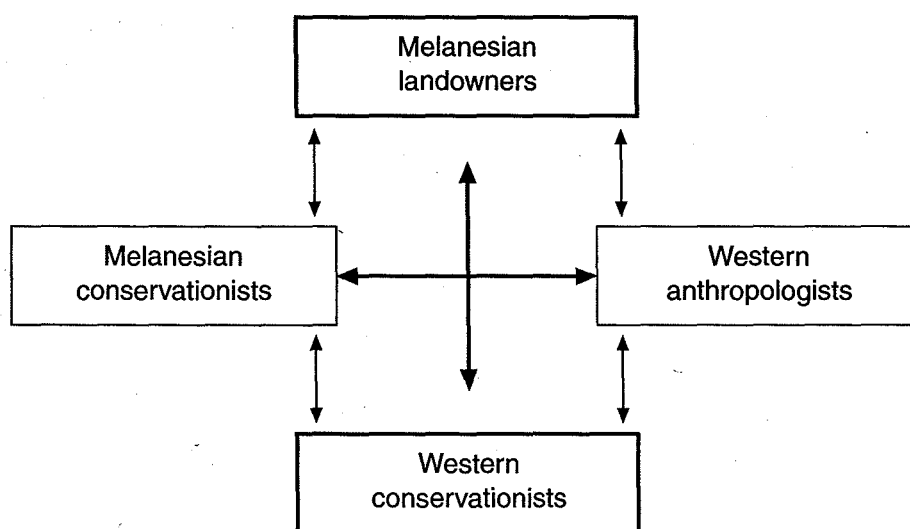


Figure 5.2 The characters engaged in talk about the value of indigenous knowledge.

How does the topic of indigenous knowledge actually enter the construction of this play within a play, and how does its appearance affect the construction of lasting 'conservation covenants' between the two main parties? To address this question, I describe a few scenes from this play with which I happen to be familiar as a result of my own participation in the conservation policy process over the course of the past decade. From my reflection on these snatches of dialogue, I shall then try to extract some messages that might serve to advance the terms of the debate between anthropologists who have an interest in conservation and conservationists who have an interest in anthropology.

The road to hell is paved with logs

The latest phase in PNG's conservation policy process began in April 1990, when the World Bank orchestrated a round table meeting that gave birth to a creature that was eventually called the National Forestry and Conservation Action Programme. One of the few unexpected outcomes of that meeting was a resolution to establish a task force on Environmental Planning in Priority Forest Areas, with a mandate to take immediate action to counter the threat of logging in areas which were thought to contain plentiful amounts of biodiversity. The Task Force spent most of its initial donor funding on a couple of expeditions to potential conservation areas in Milne Bay and New Ireland provinces, and in each case, an anthropologist was sent along to reflect on the feasibility of achieving 'formal landowner acceptance' of a conservation strategy which had not yet been devised (Filer 1991b; Young 1991).

And so it was that three national stakeholders and I descended (literally, by helicopter) on the people of the Lak census division, at the southern extremity of New Ireland, in August 1990. Our first port of call was the home of the locally elected member of the New Ireland provincial government, and the first of our three public meetings was held that night beneath his house, as the rain poured down around it. Our host warned us in advance that his own personal interest in protecting the local environment was not shared by most of his constituents, and that his own survival as an elected politician depended on his ability to represent the views of those people who 'pull the strings on the backs of us leaders'. If he was referring to the village elders who later joined the discussion, then we could understand why he had been persuaded to facilitate the deal between a Malaysian logging company and the local landowner company that was already awaiting the imminent approval of the relevant national government ministers. These old men could only talk of how much they had 'suffered' from the experience of living in the 'last' (most backward) corner of the province, if not the whole country, of their immediate need to secure some small amount of money to ease that suffering before they died, and of their absolute determination to sell their trees in order to get it. Their brains were pained by any thought of an alternative.

And what was the alternative? Not having a precise answer to this question,

my colleagues and I did our best to improvise a floor show that contained a number of compelling images without making any specific promises. Talk about laws, policies, permits and agreements would surely lower the local brain pain threshold to an unacceptable degree. Our time was short, and so was the patience of our audience. So we talked about roads instead, not only because the logging company was promising to build them, but also because the Tok Pisin word *rot* provides the metaphorical key to almost every thought about development in rural Melanesia. And we talked about the Bible as well, because this was the most familiar source of parables and homilies on such topics as poverty, greed, and deliverance, and hence, perhaps, on more elusive concepts such as 'conservation' and 'sustainable development'.

In our exposition of the two-road theory of development, those who travel the logging road, which looks so wide and straight at first, will sooner or later find that their progress is cut short by fallen trees, and that the road itself does not last very long, as logging roads and logging royalties both run out very fast in PNG. So this is the road to hell, or maybe a road that goes round in circles, and therefore does not get you out of the last corner. The better road is the long and winding one, where the long-term benefits are greater, but so are the short-term sacrifices. The last people to make the choice could be the first people to choose the right road, and the first to discover the delights of the 'benefit package' with which the international community (or maybe God) was preparing to reward them.

In all three public meetings, we found that this kind of semi-Biblical talk had greater resonance than some of our other spontaneous experiments in landowner awareness. For example, our audience saw little merit in the suggestion that future generations of local landowners might regard the decision to sell a lot of trees to an Asian logging company in the same way that the present generation regarded the earlier sale of land to European colonizers for the price of an axe or a piece of cloth. On the other hand, our talk about 'another road' provoked their recollection of the many empty promises which the government had previously made, and our own association with the national Department of Environment and Conservation might well be taken as evidence that we were just the latest in a long line of hot air merchants.

One member of our team tried to overcome this obstacle by talking about a rare species of butterfly, which, according to his estimate, could be captured and sold for 50 kina (then worth more than 50 US dollars) a piece on the world market. In order to reinforce the point, he captured one himself and carried it from one meeting to another, using it as an illustration of the potential economic value of an unlogged forest. But that butterfly was the only visible and concrete thing we had to offer. The rest was just talk, and might mean nothing, as members of our audience constantly reminded us. We argued that the logging company had also made many promises that might prove to be false, and that the loggers had more reason to deceive the people than we did, because they stood to make a profit out of their promises, whereas we did not.

But the loggers had already provided benefits, including quantities of money, which had much greater value than the sight of a butterfly and a lot of talk about another road.

The Task Force decamped from Lak with a vague promise to return with a map which would represent the compromise proposed by the provincial member at our first wet meeting – that a way might be found of dividing his electorate into areas which could be logged and areas in which the ‘other road’ could be pursued. But my own faith in the feasibility of such a deal had been thoroughly shaken. Shortly afterwards, I wrote a newspaper article in which I made the following observation:

Our conservationist friends in the developed world sometimes seem to imagine that Papua New Guinean villagers resemble the Indians of the Amazon rainforest – people living in simple harmony with Mother Nature, for whom ‘development’ is a menace imposed by outsiders. If these people went to Lak, they would be in for a big shock. It would probably be difficult to find another place in PNG where local landowners were more insistent on the need to have their trees cut down as soon as possible!

(Filer 1991a)

By the time that this article was published, in January 1991, the Minister for Forests had already issued a timber permit to the local landowner company, and the cause appeared to be lost. The Task Force never did return to Lak, nor did it ever go anywhere else, so the scepticism of our local audience would seem to have been justified.

But this was not the end of the local story. In August 1992, long after the task force had been disbanded, the provincial politician sent a letter to the International Tropical Timber Organization, asking for help to control the environmental damage caused by the logging operation which was now under way, and to establish a ‘Conservation Foundation’ and a ‘Wildlife Management Programme’ incorporating profitable environmental activities’. This letter eventually found its way back to the desk of a New Zealander who had just been appointed as Chief Technical Adviser to a new Biodiversity Conservation and Resource Management Programme which was to be housed in a Conservation Resource Centre attached to the national Department of Environment and Conservation. This entity had just been granted five million US dollars by the Global Environment Facility, primarily for the purpose of establishing a pair of experimental ‘integrated conservation and development projects’. The Western conservationists associated with the programme read the member’s letter as ‘a social invitation from a unified group of landowners eager to explore alternative methods of forest development’ (McCallum and Sekhran 1997: 19). So they spent the next three years in what eventually proved to be a losing battle with the loggers and their local allies. This seems somewhat strange, in retrospect, because the Chief Technical Adviser and his national counterparts already had copies of my own

detailed report on the activities of the Task Force (Filer 1991b), which certainly did not support their chosen reading of the member's letter.

One can only conclude that their decision to return to Lak was either the result of bureaucratic inertia and structural amnesia, or else a deliberate move to 'take the fight to the enemy' by taking the local politician's words at face value, and seizing the opportunity to mobilise a re-enlightened community against the forces of darkness which it had previously been so eager to embrace.

(Filer and Sekhran 1998: 248–249)

At any rate, the lessons learnt from their first big experiment (McCallum and Sekhran 1997) confirmed at least three of the lessons learnt in the brief encounters of the Task Force: the best (if not the only) way to persuade landowners that conservation is a good thing is to represent it as an alternative form of development; it is very difficult to represent conservation as an alternative form of development without raising expectations or making promises about the delivery of such development; and it can also be very difficult to engage landowners in a dialogue about conservation and development options without exaggerating the intensity of factional struggles within their communities.

The excavation of rural desire

The steadily unfolding failure of the Lak experiment did at least cause its conductors to take more interest in the stories told by anthropologists about the people who live with lots of biodiversity in their backyards. When the Conservation Resource Centre decided to initiate its second project in Madang Province, on the northeastern slopes of the Bismarck Range, I was invited to join the first patrol to the 'area of interest' in April 1995 (Filer *et al.* 1995), and then to assist in the design of a social feasibility study which was to form an integral part of the 'Bismarck-Ramu project'. By the end of 1995, the lessons of Lak had been absorbed into a framework plan for the new project, which asserted that:

ICAD community liaison staff must start by not talking about conservation, not talking about economic incentives, and not talking about development. . . . Instead they should be listening to the community, and facilitating a debate within the community as it explores its own needs and beliefs . . . Conservation and the establishment of protected areas will be the outcomes, not the motivators.

(quoted in Ellis 1997: 16)

If conservationists could not hope to outbid developers for access to natural resources on customary land, they should stop thinking about conservation as an opportunity cost for which landowners ought to be compensated, and start

thinking instead about the pursuit of something called 'genuine community conviction'. And in the Bismarck-Ramu case, it looked as if they had the time and space to test the weight of such moral incentives, without having to counter the material incentives offered by immoral developers.

This strategic innovation coincided with the appointment of a new Chief Technical Adviser to the Biodiversity Conservation and Resource Management Programme. Unlike his predecessor, whose international career had been launched on the back of his experience as a park ranger in New Zealand, the new man was an applied anthropologist whose main qualification was his experience of working with an Aboriginal Land Council. Unlike his predecessor, he also knew a good deal about things like Participatory Rural Appraisal (PRA), and might even have shared the responsibility for insisting that all the stakeholders now mixed up in the Bismarck-Ramu project, including himself, should participate in a workshop on this subject in order to refine a new form of 'community entry'. The PRA Toolkit Workshop was duly held in May 1996 (Grant 1996), and the Bismarck-Ramu project began its new lease of life as an exercise in the discovery of self-reliance.

The workshop served as an initiation ceremony for the new members of the Community Development Team that was to test the level of community conviction in the designated Area of Interest. The second stage of their initiation was a training programme orchestrated by an American Community Development Trainer who had a long track record of awareness work in PNG village communities. His own written records of this training exercise feature the deconstruction of two texts produced by other Western conservationists associated with the project. The first of these was an essay written by an American entomologist who now doubled as the project's education coordinator. This essay (Orsak 1997) took the form an imaginary dialogue between conservationists and landowners, in which the latter presented a familiar litany of reasons for not listening to the former, and the author suggested a number of ways in which the conservationists might rephrase their arguments in order to avoid such reactions. One of the points made in this essay was that traditional Melanesian culture lacks the sort of 'conservation ethic' that might encourage landowners to accept the arguments of conservationists in a modern political setting.

But members of the Community Development Team were reportedly unable to understand, or unwilling to accept, this argument. They felt that Melanesians were accustomed to think of themselves as being part of their natural environment, rather than being separated from it, and this meant that there could be no point in making a distinction between the idea of people looking after their forest and the idea of 'bush spirits' (*masalai*) looking after it, or even the idea of the forest looking after itself (Lalley 1998). The Community Development Trainer took this to be a sign of the gap which really exists between Western and indigenous conceptions of the natural environment, rather than a function of the role which the trainees were now expected to play as mediators in the

conversation between conservationists and landowners. The trainees themselves were disposed to expect that indigenous knowledge and community conviction would be found together in the bed of self-reliance, once they got beneath the mental blankets of 'development-dependency'.

The second text whose deconstruction helped to shape their vision was the opening speech that project staff had made at village meetings during previous patrols to the Area of Interest. The result of this exercise was a prototype or paraphrase for the opening speeches that would henceforth be made by members of the Community Development Team:

We are not working for a bank, a logging company, a mining company, a church, or a political party. We are not even working on a 'project', we are simply an 'environmental group' (*environmen lain*) working with the Department of Environment and Conservation. We are not going to help you to 'conserve' or 'develop' your 'resources', nor is the 'government' going to do so. We are only here to talk. We don't know what you know. Perhaps we can teach each other something. Perhaps we can help you to help yourselves. And we will only stay here if you want us to.

(CRI 1996a, Appendix 3)

This display of verbal diffidence was intended to prevent the listeners from engaging in what the team members later came to describe as a process of 'jumping through hoops' in order to capture the windfall benefits which villagers have learnt to associate with passing projects. But it also contained an element of bad faith in its own right, because it concealed the ultimate objective of a donor-funded conservation project, as if this were a secret mystery which could only be revealed to those villagers who learnt to jump through another set of hoops in order to demonstrate their 'conservation conviction'.

The Community Development Patrols were sent out at intervals of approximately two months. In each case, the Community Development Team spent a few days in a joint 'briefing session' under the supervision of the Community Development Trainer, and was then divided into two or three smaller groups, each spent about three weeks visiting a number of local communities in the area of interest, before finally returning to base for a 'debriefing session' in which their findings were recorded for posterity. The 'community development process' was itself divided into seven steps, each of which required a separate visit to any given community (Ellis 1997: 21). During the first two visits, the interaction between team members and local villagers was conceived as a process of 'story-telling', 'trust-building', and the 'hearing of community themes' (ibid: 30). The more elaborate instruments in the PRA Toolkit were saved for the 'diggings' undertaken in subsequent visits, lest they be construed as 'gimmicks' by an unfamiliar audience (van Helden n.d.).

After eight rotations of the fieldwork cycle, 34 communities had received at least one visit from the Community Development Team, but very few had

reached Step Five, which was the point at which they were able to formulate an 'Action Plan' to address their own problems in a self-reliant manner, and hardly any had ascended to Step Seven, at the very top of the ladder, where the role of project staff would supposedly be limited to monitoring and facilitating the implementation of this plan. That is partly because most of the communities had received less than five visits by the end of 1997, but in some cases, the process had been cut short by the failure or refusal of local villagers to play the game according to the rules developed in the briefing and debriefing sessions which took place backstage.

Backstage was the place where nearly all the Western conservationists associated with the project stayed throughout this period, in order to avoid being cast in the role of 'bosses' whose white skins might be taken to presage the delivery of 'cargo' (CRI 1997b: 6). The only significant exception to this rule was the Social Feasibility Consultant, a Dutch sociologist attached to my own division of the National Research Institute, and even he was not allowed to enter a community until its members had safely climbed the first two rungs on the ladder of self-reliance. His role was to document the findings of the barefoot anthropologists in the Community Development Team, to verify or supplement these findings through his follow-up patrols, and then present the outcome as a descriptive account of social relations in the Area of Interest (van Helden 1998a) and a more prescriptive assessment of the 'social feasibility' of establishing a conservation area within it (van Helden 1998b).

Two fundamental 'community themes' were dug up from this field of inquiry. One was the relationship between thousands of 'Jimi people', including those resident in the upper Jimi Valley and those living on the margins of the Ramu floodplain, and a few score 'Ramu people', most of whom were living along the banks of the Ramu River. At one stage, members of the Community Development Team described this relationship as an instance of 'neo-colonialism', in which the Jimi people were colonizing the Ramu people in both a physical and cultural sense, as most of the 'original landowners' had already begun to speak the language of the newcomers who were taking their land and plundering their resources, and the highlander's fear of lowland sorcery had become their last line of defence (CRI 1996b: 45). The second theme was the intense political rivalry, and occasionally violent conflict, between the clans and sub-clans into which the Jimi people were divided at all altitudes. For those who formed the colonizing frontline in the Ramu floodplain, these political relationships were further complicated by the fact that a minority did have a solid claim to be the traditional owners and occupants of the Bismarck foothills, and even segments of the floodplain itself, while the majority did not (van Helden 1998a: 236-239). In this social and political context, the Community Development Team made the somewhat unremarkable discovery that some of the Ramu people had an interest in the conservation of indigenous knowledge and natural resources, while most of the Jimi people were primarily concerned with the distribution of wealth, status and power.

In retrospect, the weight of demands with which the Jimi people ultimately crushed the community development process might be blamed on the expectations which had already been raised before that process was initiated, and might thus be taken to confirm the principles by which it was designed. In the Jimi Valley, the Community Development Team was almost ambushed by a gang of youths who thought it was a 'group of company officials or agents' bearing large quantities of money (CRI 1996b: 27-28), and was later confronted with demands for compensation for the 'theft' of natural resources by the 'foreigners' who had conducted a biological survey in October 1995 (CRI 1996c: 37). And on the Ramu floodplain, the team was constantly embroiled in a continuation of the squabbles about payment and employment that had first erupted when the project base was set up next to the half-built airstrip back in February 1996 (CRI 1996b: 30-31; Ellis 1997: 29). But it soon transpired that the root cause of such misunderstandings and disputes could be traced back to the employment of the airstrip's local architect, a university graduate, as the project's Community Facilitator, and their subsequent escalation was due to his own determination to 'capture' the project in order to win votes at the national election which was due to be held in June 1997.

The Community Facilitator was duly summoned to headquarters and placed behind a desk, and therefore had to quit the project to continue his electoral campaign. But his method of advertising the project's existence still served to undermine its objectives. His claim to be the 'boss' who had secured a huge amount of project funding from the World Bank (CRI 1997b: 16) caused all sorts of rumours to invade and obstruct the Jimi people's participation in the community development process. There were some who partly believed his claim, and wanted to know why they were not getting a bigger share of this 'free money'; there were others who were somewhat sceptical, and wanted to know when they would get to meet the 'real boss'; and for those who did not trust this candidate at all, there was the option of believing that what he had really done was to help the government sell their land to the World Bank (CRI 1997a: 10, 15, 42). Those members of the Community Development Team who hit this wall of ideas in the Jimi Valley were also warned that their lives were in danger, so they beat a hasty retreat.

The concept of the project did not enable the excommunicated community facilitator to win the election. But even after the votes had been counted and the rumours had subsided, the Community Development Team was still lamenting its inability to wean the Jimi people away from their 'handout mentality'.

The team's observation of the attitudes and body language of people during meetings gave an indication that there is really little interest in any community development if it requires the people to get organised or do actual work. The community has said again that, 'The CDT come and come again and nothing is given to the villagers'. It appears that in spite of

the amount of time put into the village development process, the communities still have an overwhelming reliance on 'cargo' and they are waiting for the CDT to bring something.

(CRI 1997b: 39)

The persistence of such 'cargo expectations' was not only linked to the many promises which had just been made by candidates competing in the national election, but also to the promise of drought relief which followed shortly afterwards, to the experience of an influenza epidemic which also afflicted the area in 1997, and to the more abiding shortage of both natural resources and development opportunities in the Jimi Valley (Ellis 1997: 43; van Helden n.d.). But when one reads the reports of the Community Development Patrols, one finds that the number of conversations in which the Jimi people wanted to know what the team had to offer is matched by the number of conversations in which they wanted to know what the team really wanted. For example:

They always want to know exactly what we are after. Their never-ending [enquiries] make us feel like we have confused them more than ever by coming in and working with them ... [They] keep coming and asking questions such as: 'After you had visited us for so many times and have had enough of that, what will come next? What are you going to do; what are you going to establish? ...'.

(CRI 1996b: 41)

And it sounds as if they never could get a straight answer to such questions.

The talk presented by the Ramu people was considerably less demanding, but in some ways more mysterious, less political, and more religious in its flavour. While most of them had not been 'spoilt' by the sight and sound of Western conservationists during the previous phase of the project, the concept of the 'cargo cult' was no less evident in their reactions to the process of 'community development'. But while the Community Development Team used this concept to explain the Jimi people's failure to realize or practise the virtues of self-reliance, the Ramu people used it to explain, and in one case to reject, the process of discovering these values. In other words, the Ramu people thought the project which was not a 'project' was perhaps a cult, whereas the Jimi people knew the project was a 'project', or subscribed to the belief that it must be a 'company' of some sort, and demanded that it yield the sort of benefits which members of the team called 'cargo'.

When the Community Development Team made its second visit to one Ramu river hamlet, the headman was quite effusive in his greeting:

My dear children, I am greatly delighted that you had cared enough to visit me again as you had promised me the last time. What better time could you have chosen to visit me and your brothers and sisters here, than during

such a time like this when I am no better than dead. . . . You must NOT leave us, never, remember that we are family now. We are on the ground now; we haven't yet set foot on the first step on the ladder leading up to that house you mentioned. And the door into the house like you mentioned is way up there out of our reach. Just think of the taro. When you plant it, does it grow, mature, bear and ripen overnight? Not that I know of! Or do you know of such a 'miracle' taro? My children, let's talk reality and stop dreaming! In the same way as we wait for the taro we plant to grow, mature, bear and ripen; you and I (us) must work hard and take our time. We must NOT rush, for rushing is risky, as far as my experience makes me believe.

(CRI 1996b: 42)

One of the headman's sons glossed this welcome by declaring that the team had been brought to them by Jesus Christ himself, and made allusions to the parable of the sower and the seed:

You are the sower and your words are the seed. We who listen, for our part, are the different types of soil. What we do after we hear your words and even after you had gone will show what kind of soil we each are. If we are good soil then the seeds you sow will grow and bear fruit.

(ibid)

Encouraged by such thoughts, the team set off to make its initial visit to a neighbouring hamlet, whose inhabitants were also 'surprisingly hospitable' and exceedingly devout. On the other hand, their spokesman's response to the team's explanation of its 'work' was quite at odds with their initial hospitality:

My people and I have the right to know whoever comes into our land and also why he or she comes in. We asked you to clarify to us your work and you claim that you have done so. The fact is, we are even more confused, because your words are like complete nonsense, they are absolutely meaningless. . . . How can we possibly make our decision/stand if we are not clear about your work? Therefore we ask you not to come again here in the future . . .

(ibid: 44-45)

So the team members said 'thank you very much' and quit the scene.

They were understandably puzzled by this negative outcome, which was apparently unique in their experience of first encounters. But a partial explanation emerged during a subsequent patrol along the banks of the Ramu River, when the team members got a 'very cool reception' in two other hamlets, including the one whose headman had previously spoken of them as his children. Now the headman addressed them as 'spirits of the dead people' (CRI

1997a: 21), and they were given to understand that the community which rejected their initial advances had since

spread rumours about the CDT's being members of a cargo cult known as 'Bembe'. The suggestions were that we were spirits that had come back from the dead (*daiman*) and sinners (*sinman*) and that people would fall ill if they spent the money that we had paid them for accommodation. People were obviously very confused because last time the CDT's had come and when somebody had been ill, they had given them medicine and had treated him well. People initially did not know whom to believe and waited without giving us food until they had heard our side of the story. It seems that our exercises, including going back and forth to town and resource mapping had backfired on us in the sense that local people had interpreted them as part of the magic that we were using to bring them under the spell.

(ibid: 42)

But the headman himself, who seems to have believed this story, still wanted the team to make regular visits, 'otherwise the communities would lose interest in us' (ibid: 21). The team members thought it was 'quite ironic that the strong emphasis on avoiding unrealistic expectations and self-reliance makes people think we are a traditional cargo-cult', but they also found something more rational and sinister in the hostility of one local leader who was 'strongly in favour of logging and had rightly identified us as potential opponents to such activities' (ibid: 42-43).

Whatever their real motivation, those Ramu people who espoused, or were said to espouse, the ironic perception of the project as a sort of cargo cult did not reach the top of the ladder which led to the 'house of conviction'. The first signs of this achievement were detected amongst the small group of Ramu people encountered by the very first patrol to the Area of Interest, living in immediate proximity to a large bunch of very demanding Jimi people. And for this very reason, their convictions had a question mark against them, because the community workers and project managers could see them 'not as a drive for conservation alone, but as a way of addressing the major social problem of integrating the Jimi settlers, with their aggressive land use patterns, into the Ramu way of life' (Ellis 1997: 49-50). The minutes of meetings held at the end of 1997 suggest that it might not have been a 'drive for conservation' at all. When the Community Development Team met with members of the two landowning clans, and told them that 'the Project would only be involved with them if they want to conserve their natural resources', the landowners evaded this question by asking that 'the team should not meet and work with the Jimi settlers' (ibid: 56). The team disregarded this request, and held a separate meeting with the settlers, in which they rehearsed the substance of their earlier discussion with the landowners.

The settlers agreed and supported the landowners' idea of conservation. They reported that there were rumours that some of the landowners had interests in logging. . . . The settlers see that if logging is brought in they will be adversely affected. Each family has a very small area to spare for logging. Secondly, the areas that they normally hunt in which are beyond what they were allocated by the landowners might be destroyed by logging. They will find it very hard to continue living the way they do now. Hunting and gardening practices would be affected.

(quoted in Ellis 1997: 56)

When the Conservation Area Manager finally came to talk to the landowners in April 1998, they had already reconstructed their desire for conservation as a desire to make provision for the livelihood of 'future generations', thus neatly aligning themselves with the Fourth Goal of the National Constitution, but this was still not enough to convince the conservationists of their sincerity (*ibid*).

Greater progress seems to have been made in meetings with the residents of another village on the Ramu River, even though their interest in conservation also stemmed in part, if not entirely, from their wish to keep or get the Jimi people off their land. This was the village where people could clearly remember the original visit by an officer of the Australian colonial administration

who discussed the establishment of a National Park and told people to limit their hunting in certain areas and not to use guns. The Ramu river communities still purport to observe these rules but are very concerned about the continuous hunting on their lands by Jimi Settlers. It is not entirely clear whether they understand the full concept of a national park or whether their interest in such is mainly a result of the need to stop the Jimis from further encroaching on their ancestral lands.

(quoted in Ellis 1997: 51)

These people had reiterated their interest in the establishment of a 'national park' when a team of government officials came back to assess the original proposal in 1986 (Filer *et al.* 1995: 47), and repeatedly brought the matter up in conversations with the Community Development Team. Indeed, one might well wonder what difference, if any, the team actually made to these people's stated preference for 'conservation', apart from persuading them that the government would not do very much, if anything, to help them realize their goal. It is true that they had to prove their own capacity for self-reliance by meeting the cost of a trip to Madang to meet the Conservation Area Manager, and when this action was rewarded by his return trip to the village, the villagers had found some more 'convincing' reasons for their preference. Apart from defending their resources against the encroachment of the Jimi people, they also wanted to reinforce clan boundaries within their own community, they were worried about the possible impact of a large-scale mining project, and they saw

some potential for attracting tourists to the area. But when the Conservation Area Manager raised the question of whether traditional beliefs and practices would provide a suitable basis for conservation, one villager replied as follows:

We cannot use the same practices used by our ancestors nowadays, because educated people do not respect and listen to the village elders. Educated people are proud of themselves, they think that they have been to school and are more knowledgeable than the village elders. Therefore if village elders make rules to conserve a certain area, people that have some form of western education will not adhere to those rules.

(quoted in Ellis 1997: 57)

This sentiment did not exactly fit the bill to which the Community Development Team was wedded at the outset, which was to find or plant 'community conviction' in the rediscovery and reproduction of indigenous knowledge.

At least the Ramu people showed a lot of interest in the subject of indigenous knowledge, even if they were sometimes inclined to reject it as the work of the devil. But the manner in which the subject was broached in their conversations with the Community Development Team would only have confirmed the Jimi people's prejudice against them. On one occasion, as they set off to visit a Ramu community, the team members received the following piece of advice from their Jimi hosts:

They will not feed you, they will not house you. And if they ever do house you, this will mean you sleeping underneath their houses. They don't even have food anyway, for they know no gardening. We were the ones who taught them gardening and all other skills; we civilized and 'tamed' them. They live only on sago. They can't even talk sense with you like we do, and they will never understand your message.

(CRI 1996b: 41)

What made 'sense' to Jimi people, by and large, was talk about 'development', which members of the team refused to talk about because they thought that it was just another name for 'cargo' (Ellis 1997: 36). And what did not make sense to Jimi people, by and large, was the team's inclination to talk about indigenous knowledge as knowledge about 'the bush', about nature rather than culture, and to talk about its value in spiritual, rather than purely practical, terms.

Yet I would not go so far as to say, from the evidence available, that the Jimi people had their feet firmly planted on the ground, while the Ramu people had their heads in the clouds or their minds in 'the bush'. On both sides of the fence, villagers were equally concerned to search out ways of establishing material exchange relationships with the Community Development Team, and seem to have been equally puzzled by the team's reluctance to reciprocate (CRI

1997a: 11; van Helden 1998a: 258). And it would be hard to argue that one side had more or less of an interest in the prospect of large-scale resource development, whether by logging, mining or petroleum companies. In this respect, the difference resided in the fact that such companies really did show rather more interest in the resources claimed by the Ramu people than in those claimed by the Jimi people, and if the latter really did believe that an Iraqi logging company was about to harvest the timber in the upper Jimi Valley (CRI 1997b: 15), then theirs was by far the greater delusion.

The bottom line, if such it can be called, is that the people who were most convinced about the need for conservation were Ramu people who harboured a desire to roll back the tide of territorial encroachment by groups of colonizing highlanders, and were much less concerned about the threat posed by foreign resource developers (van Helden n.d.). Proponents of the project's community development strategy would say that it does not matter why people want to conserve their resources, so long as they really want to do it (Ellis 1997: 54). But then we have to ask whether this conviction stems from the correct application of a PRA Toolkit, and whether those who are convinced have also got the power to achieve their goal when they have learnt that no one else is going to help them if they cannot help themselves. However long the Community Development Team spent 'digging' into the local social soil with its 'timelines' and 'resource maps', what it found at the bottom of the hole was something already present on the surface – a set of disputed territorial boundaries and unequal spatial relationships.

Selling the biodiversity business

The Conservation Resource Centre was in fact a rather peculiar creature in the conservation policy process precisely because it was housed in the national Department of Environment and Conservation, and therefore had the benefits and costs of being seen as part of the 'government' that normally failed to meet the expectations of rural villagers. Most of the other ICDPs in PNG have been designed and implemented through partnerships between national and international NGOs, whose own specific aims and interests have variously coloured the nature of their dealings with local landowners. Three of these projects received substantial financial support from an organization known as the Biodiversity Conservation Network (BCN), whose approach to the business of establishing conservation areas in PNG was markedly different to that which had recently been espoused by the Bismarck-Ramu project. These contrasting strategies or scripts alert us to a range of possibilities for local variation in the substance and outcome of the talk which has been going on between local landowners and Western conservationists.

The BCN was initially established as a five-year programme in late 1992, but the period of implementation was later extended to six and a half years, which meant that the programme came to an end in the first half of 1999. Its funding was actually treated as part of the US government's 'attribution' to the Global

Environment Facility (BCN 1997: 120), but it was implemented as one component of the more enduring Biodiversity Support Program, which is sponsored by a consortium of three American NGOs – the World Wildlife Fund, The Nature Conservancy and the World Resources Institute. During the six and a half years of its existence, the BCN provided implementation grants to a total of 20 conservation projects in the Asia-Pacific region, in order to evaluate the effectiveness of what were described as ‘enterprise-oriented approaches to community-based conservation of biodiversity’. This meant testing a ‘core hypothesis’ that stated that ‘if local communities receive sufficient benefits from an enterprise that depends on biodiversity, then they will act to counter internal and external threats to that biodiversity’ (ibid: 1).

The various NGOs that were on the receiving end of these grants were apparently required to volunteer some of their staff for a form of initiation into the mysteries of the project cycle (that ceremonial practice which is familiar to denizens of the aid industry) in order to improve their capacity for ‘adaptive management’. The initiates were thus taught to rethink their projects (and now also think about how to spend the BCN grants) by following the path that leads from the production of a mission statement to the process of revising management and monitoring plans in light of lessons learnt from their implementation (Margoluis and Salafsky 1998). The second stage of this initiation involved the production of a ‘conceptual model’, which consisted of a lot of boxes containing the various factors which had some positive or negative impact on the conservation of biodiversity in the place where a given project was located, and a smaller number of circles containing the specific project ‘interventions’ which were intended to modify the causal connections between these factors, and thus to test the core hypothesis.

In January 1999, as the BCN programme drew to a close, Network personnel convened a public meeting to reveal the lessons which had been learnt from their experiments with the core hypothesis, and to assess the extent to which various threats to biodiversity had actually been diminished through the pursuit of each project’s enterprise component. After the newly anointed Melanesian Masters of Business Administration had recounted their own experiences in the biodiversity business, their patrons in the Network offered the audience a number of ‘general but non-trivial guiding principles’, which took the form of ‘conditional probability statements’:

For example, we might say ‘In Melanesian type social systems, it is generally better to work with the big man to solve conflicts unless he is corrupt’. The key features here are that the principle applies to more than one place (in Melanesia) but not everywhere. Furthermore, it is not guaranteed to work in all instances – the user has to be smart enough to apply it to his or her own situation – for example, to determine if the big man is corrupt or not.

(BCN 1999: 6)

Two particular points struck me as I listened to this presentation. One was the evangelical zeal of the American conservationists associated with the Network, which had presumably permeated all their previous conversations with the Melanesian conservationists whom they had been initiating into the business of turning biodiversity into a commodity. The other was the fact that local villagers were represented, by the Melanesian conservationists, as people who still had a great deal to learn about this new form of entrepreneurial culture, and who still had to be persuaded to abandon a variety of economic practices which were themselves regarded as major 'threats to biodiversity'. If the Melanesian conservationists had been preaching with the same vigour as their American mentors, and thus breaching one of the main taboos imposed on the Bismarck-Ramu project, they had not yet won too many converts from the communities that hosted their projects.

My only contribution to the public action in this scene was to ask one of the national staff employed on the Lakekamu Basin project why he thought that small-scale alluvial gold mining was such a big threat to biodiversity, and how he expected the local villagers, who had obtained the bulk of their cash income from this activity for many decades, to sacrifice it in exchange for money to be earned from ecotourists who had not yet put in an appearance. His answer to the first question was that people who practice small-scale alluvial mining are more likely to welcome the prospect of a large-scale industrial mining project than people who do not. His answer to the second question was that it was very difficult.

The small miners in question had originally learnt their craft from the European prospectors who were active in this area during the second and third decades of the twentieth century, and who then vanished from the scene (Nelson 1976). So their knowledge of mining was not exactly 'indigenous', even if it was by now customary, and since 'we all know' that mining is a dirty business, it was evidently not the sort of knowledge which had any sort of place in an 'entrepreneurial culture of conservation'. But 'we all know' that logging is a dirty business too, especially when it is done in 'virgin tropical rainforests'. So if Western conservationists feel the need to wean indigenous miners off their gold pans in order to save them from opening their arms to large-scale mining companies, how can they also be selling the virtues of walkabout sawmills to villagers who might otherwise open their arms to large-scale logging companies? For that is what they do in many other Melanesian conservation projects, where knowledge of small-scale saw milling is not even traditional, let alone indigenous.

No one at the meeting had a ready answer to this final question. Perhaps the answer lies in the physical or conceptual difference between rocks and trees. Eco-forestry adds economic value to the forests which contain biodiversity values, and if we get our definitions right, it simultaneously causes its practitioners to appreciate the value of the biodiversity in the forests which they are only very gradually cutting down. Mineral resources have no place in such equations, and the pursuit of gold, by any means at all, is nothing but a testament to human folly.

The gold-panners of the Lakekamu Basin had not yet had much opportunity to evaluate this line of reasoning, because the new business on offer was ecotourism, not ecoforestry. Or perhaps it was no new business at all. As one of the national conservation project staff described the prospect:

After the first year, we had a training session about ecotourism. One of the guys got all inspired and wanted to build a guesthouse. He organized his family and built the place. This was really hard. I was glad that he was showing interest, but I was worried about not having any guests come. They built the house and then they started asking when the tourists would come. I didn't know what to tell them. And this has turned out to be a problem. Only a few people have come and already the guesthouse is starting to fall apart. I feel responsible for what has happened and that I let them down. Even now, when I go back to the Basin, they ask me, 'When will they come? Is there any news of tourists coming?'

(BCN 1999: 166-167)

While this family waited for Godot, the alluvial miners, crocodile hunters and betelnut farmers went on minding their own business, mindful perhaps of other promises and prospects of 'development' which had previously come to nought.

Back in 1988, I conducted a social impact assessment of a medium-scale industrial mining project in this area, in which I made the observation that

it is better to promote those money-raising activities which have already proved viable, and even those which have previously failed for identifiable and remediable reasons, than to promote novel activities whose viability is an unknown factor.

(Filer and Iamo 1989: 58)

The project in question never got past the design stage, but I certainly did not find that this prospect of 'development' held any special appeal for the alluvial miners in the area, who had tried and failed to obtain the mining company's assistance in getting their own product to market, and who were understandably concerned that it might later find some way to put them out of business. Little did they know that, when the mining company had gone, a conservation company would come and look for ways to do so.

Disintegrating conservation and development

I have chosen to represent the conservation policy process as a play full of odd characters and strange talk, not because I wish to emphasize the absurdity or futility of that process, but because I want to underline its unpredictable, disputed, open-ended quality. Some anthropologists might prefer to deconstruct the narratives contained in that tedious mass of project and policy documents

which is produced by global actors like the Global Environment Facility (Zerner 1996). But this kind of discourse analysis only seems to reinforce the semblance of imperial power that it purports to criticize (Grillo and Stirrat 1997). The discourse of 'integrated conservation and development' actually disintegrates in talks between the 'stakeholders' who do not live in Washington and do not read academic journals. It is only by participating in these conversations, both as listeners and speakers, that we can see how conservation policy is manufactured on the national and local stages where the global script does not dictate the outcome of the play (Croll and Parkin 1992: 33-34).

Many anthropologists have made it their business to debunk what Ellen (1993: 126) calls the 'myth of primitive environmental wisdom', and Melanesian specialists (myself included) have been amongst the most vociferous exponents of this stock in trade (Bulmer 1982; Dwyer 1982, 1997; Allen 1988; Filer 1991b, 1994). But what concerns me here is not so much the truth as the politics of this sceptical approach to the value of indigenous knowledge in the conservation business. Brosius (1999) has recently suggested that the deconstruction of essentialized images serves no useful purpose if it only demonstrates our own exclusion from the policy arena, and might, in certain circumstances, offer extra ammunition to the forces of darkness. Instead of quibbling while Rome burns, we need to ask how anthropologists can help to fight the fire.

The Melanesian version of Rome does not resemble the Malaysian political setting in which Brosius has situated his own argument, even if it does contain Malaysian logging companies. Anthropologists and conservationists alike have far more freedom to criticize the loggers and the government, and there is much less chance that anything they do or say will jeopardize the rights or compromise the strategies of customary landowners. Where Western anthropologists play many active roles in the conservation business, they may not always see the value of indigenous knowledge in the same light. Yet their views, taken as a whole, still tend to diverge from those espoused by the national conservationists employed in the same industry, who are more inclined to regard the 'myth of primitive environmental wisdom' as a rock on which to build effective rhetoric and policy.

The national conservationists have good cause to take this position, if they believe their point of view will maximize the flow of foreign funds that keeps them in employment. Occidental donors have no cause to challenge myths that help to build the confidence of their allies in the national policy community. This is one reason why the donors are prepared to pay for large numbers of national 'community workers' to be trained in the mystical arts of PRA, displacing the more rapid and sceptical appraisals offered by alien anthropologists. But the more time that is spent gathering local participation in a positive appraisal of indigenous knowledge, the greater the risk of seeming to patronize local landowners who think they need to know something more than what they think they know already (Filer and Sekhran 1998: 331). The subjects of this process have every right to wonder why outsiders of any complexion are being paid

good money for such work, when it seems to demonstrate the limits of their own access to the donor dollar.

At the same time, the concept of indigenous knowledge has gained currency in the Melanesian conservation business in a way that tends to make the adjective prescribe or circumscribe the object of the knowledge. It has to be knowledge of what and where the wild things are, because that is the knowledge that matters to conservationists who want to keep those wild things as they are. It cannot be knowledge of subsistence farming or alluvial mining, let alone knowledge of 'cargo cults' or electoral politics, because such knowledge cannot be used for this purpose. But if we cease to think of Melanesian landowners as 'rainforest peoples', and see them as they generally see themselves, as 'gardening' or 'farming' peoples, then we are obliged to recognize that the productive powers of their indigenous or traditional knowledge are weighted heavily towards the art of cultivation, and that the process of 'development' has only added to this bias.

No wonder, then, that when the community workers engaged on the Bismarck-Ramu conservation project tried to rationalize their own method of dealing with indigenous knowledge, they said they were 'digging' and 'planting' (Ellis 1997: 45), not 'hunting' and 'gathering'. But they were still looking for knowledge of the 'natural forest', and what they seem to have found is knowledge of 'bush spirits', a form of pagan religious belief, rather than the sort of ethnoscientific knowledge which could usefully be sold to foreign scientists or ecotourists. They might well argue that the separation of these two forms of knowledge reflects the alienation of the landowners from their original social and intellectual landscape, or the transformation of a disenchanted forest into an economic resource. But arguments like this hold little water with the modern Christian villager who thinks of 'custom' as the work of Satan. Since most members of the national conservation community, unlike the anthropologists and other foreign scientists with whom they work, are active members of some Christian church, they cannot readily deal with the topic of indigenous knowledge without attempting to reconcile Christian and indigenous cosmologies. This kind of syncretism might appeal to Catholics or Anglicans, but is regarded with suspicion or hostility by most of the other churches.

The designers of the Bismarck-Ramu project tried to address this problem by engaging a Church Liaison Officer to liaise with local Christian leaders who might otherwise resist the traditionalism of the community development process. This man justified the significance of his own role by commending the ability of local pastors to enter an unfamiliar community and establish their own authority within a matter of days, by simply commanding the villagers to perform essential 'public work' (Ellis 1997: 40), and certainly without spending a lot of time 'hearing community themes' or exploring indigenous knowledge. No lessons seem to have been drawn from the contrast between the commanding presence of Christian preachers and the uncommanding, non-preaching talk favoured by the barefoot conservationists. But if we compare the Bismarck-Ramu project's style of community entry with the hard-nosed business

preaching favoured by the Biodiversity Conservation Network, we may well be reminded of the different approaches which orthodox and evangelical missionaries took to the task of 'moulding Christian conviction', rather than 'conservation conviction', when they first arrived in Melanesia (Michael Young, personal communication). This does not mean that we are witnessing the spread of a new sect devoted to the worship of Biodiversity; it only means that conservationists and missionaries face a similar choice of moral incentives and rhetorical devices when they seek new converts to their cause. In both cases, the risk of appealing to indigenous knowledge is the risk of ceasing to be a 'true Christian'.

But this risk is greater for those conservationists who espouse the new Catholicism of the Bismarck-Ramu project than for those who espouse the BCN's version of the protestant ethic and the spirit of capitalism. That is because the 'Protestants' are offering material incentives, not just moral ones, integrating conservation and development by trading or exchanging the prospect of business development for the conservation of natural commodities, whereas the 'Catholics' reject developmentalism as a 'cargo cult', as yet another form of religion, and seek instead to cultivate the plant called self-reliance through the exchange of ideas, not the sale or purchase of commodities. The difference between them stems from their evaluation of the chances of successfully competing with 'developers', and most especially with logging companies, who promise more material benefits for less hard work.

The business experiments promoted by the BCN have not required a positive evaluation of traditional cosmologies. On the contrary, they look more like a set of minor rituals in the religion of developmentalism. But indigenous knowledge is still conceived as a sort of lost cause, in the sense that the current economic activities of the average villager, especially those which earn money, are seen to be minor versions of the menace posed by logging companies. The knowledge of animal behaviour that grew from traditional hunting practices has been distorted by the use of shotguns to satisfy new markets for particular species, while the knowledge of plants that grew from traditional cultivation practices has been distorted by the clearance of old forests for new cash crops (West 2000). While business is a good thing in itself, most existing forms of business are bad for biodiversity, and even the traditional arts of subsistence no longer carry the traditional guarantees of sustainability.

Yet those forms of indigenous knowledge that seem to have some bearing on the development of alternative forms of biodiversity business are also distorted through their attachment to economic activities and social relationships which are commonly less 'traditional' and less rewarding than the ones which they are meant to replace. It is one thing for the hunter to know his birds or the gatherer to know her trees, but such knowledge is not sufficient, and might not even be necessary, for people to make a decent living as small-scale saw-millers or guides on hire to adventure tourists. Nor is it obvious, either to landowners or economists, and even with substantial donor subsidies, how such occupations can generate the levels of cash income that would dissuade people from selling

their birds or their trees, or persuade a typical community of Melanesian gardeners that an unexploited forest has more value than a garden full of coffee.

Whether we take the sacred or secular road to indigenous knowledge, conservation demands more time and effort, more education and collaboration, more conversation and negotiation, than local landowners are normally willing to supply, because conservationists tend to put their projects in places which already possess a natural abundance of biodiversity values, and often little else. In circumstances such as these, the conservationists are often selling victory against an enemy who is not really present in the field, and might never get there. In which case, we might ask why any package of incentives is required to fight a phantom?

Conservationists are easily perceived as people who wish to maintain the status quo, which means poverty and a lack of services, by telling communities not to obtain 'development' by the only means that seem to be available. The populations of the Bismarck Fall and the Ramu Valley, for example, have, until now, lived under 'conservation circumstances'. They own a virtually untouched and enormous swathe of land in the Bismarck Fall, and to their great regret, experience a situation of 'conservation' rather than 'development' every day of their lives. It is unlikely that people will continue to accept this status quo if they are offered an alternative through mining, logging, or any other form of resource exploitation.

(van Helden 1998a: 255)

The conservation of biodiversity values in a country like PNG is not necessarily, or even normally, the result of talks or deals by which conservationists persuade local landowners to change their attitudes or behaviour. Where the threats posed by developers exist only in the minds of landowners as a sign of their desperation for development, it does not follow that the current economic practices of those who labour under such illusions pose a smaller or a more insidious version of this threat. Where logging or mining companies really have made an assault on biodiversity values, with or without the active encouragement of local landowners, the latter may still combine their enthusiasm for development with demands for compensation which actually serve to undermine the object of their own desire, and conservation may then be the unintended or perverse result of their attempt to capture larger rents from the developers (Filer 1994). And whether the developers are present on their land or only present in their minds, local landowners may reduce their own exploitation of local forest resources, despite rapid population growth, because they want to modernize their own lifestyles, and thus dissociate themselves from those activities, like hunting, which are hallmarks of the forest-dwellers who they do not want to be (Kocher Schmid and Klappa 1999). The moral imperatives embedded in these options cannot be readily accommodated by the contrast between a valid defence of indigenous knowledge and a blanket condemnation of development-dependency.

If conservation biologists like to think of indigenous knowledge as knowledge about biodiversity, the province of the parataxonomist who shares their own interest in the classification of wildlife, we anthropologists are more inclined to think about it as a form of cultural diversity, and even to defend and celebrate it on these grounds. But if Melanesian biodiversity values have now become a sort of metaphorical commodity on sale to the donor community, what is the current value of Melanesian cultural diversity in any but the rapidly contracting market for traditional ethnography? And how is the conservation or reduction of this cultural diversity connected with those projects and policies that are funded and constructed for the purpose of sustaining its biological equivalent? The lingering diversity of traditional culture would seem to constitute an obstacle to the conservation of biodiversity because it adds an element of uncertainty to the reaction of different local communities to the community awareness or development strategies adopted by the conservationists. On the other hand, some communities appear to regard the conservation of their own cultural identity as an activity that makes more sense than the conservation of any particular species or habitat within their territorial boundaries (van Helden 1998a: 261). Since identities and boundaries are closely intertwined, their simultaneous defence may be the only point at which local landowners can accept a modern conservation ethic as something more than a luxury which they cannot and will not want to afford until they have some more 'development' (ibid: 263).

When we think of indigenous knowledge as the property of a local community, or even the cultural unity of a culture area, we also need to remember that natural biodiversity values are inversely related to population density, and that effective conservation of these values needs a space that is normally much greater than the territory occupied by a traditional political community in Melanesia. This means that conservationists tend to end up in places which are not only 'remote' and 'backward', but which also feature traditional cultural fault-lines, often overlaid by recent population movements, where talk of boundaries is highly problematic and political, and any kind of project which distributes benefits to local people can expect to start a mass of territorial disputes. The Bismarck Fall and the Lakekamu Basin are two such places (see Kirsch 1997; van Helden 1998a). A recent gathering of PNG 'ICAD Practitioners'

noted that in the past there were no land disputes. Now that we have money there are disputes. The possibility that ICADs may exacerbate disputes should therefore be taken into account.

(Saulei and Ellis 1998: 216)

Perhaps the traditional absence of land disputes belongs in the same romantic bag as the 'traditional conservation ethic', but their diagnosis of the present situation has some merit.

While this may seem to justify the Bismarck-Ramu project's method of withholding money and promoting 'self-reliance', its resource mapping practices

could hardly fail to turn local knowledge into the knowledge of disputed territory. And if it is true that social relationships within and between Melanesian communities are normally grounded in the landscape (Filer and Sekhran 1998: 31), this may tell us more about the problem of striking 'conservation covenants' with local landowners than about their understanding of a conservation ethic or their appreciation of indigenous knowledge.

Where several local communities are given the time and space in which to make their own choice between activities which may have positive or negative impacts on biodiversity values, there is no guarantee that people who make the 'right' choice will occupy a series of contiguous territories which combine to make a single conservation area of several thousand square kilometres. If anything, it is rather more likely that neighbouring communities, or even sections of a single community, will take different options as a new means to represent and pursue their existing social and political divisions.

(ibid: 294-296)

The conservationist's dilemma here is that the weaker party generally takes the side of conservation, but the weaker party typically lacks the numbers and capacity to protect and manage a conservation area without external assistance. One of the most striking features of the current Melanesian form of landed property is that the power of local landowners to exploit 'projects' for their own ends is not matched by their power to deal with troublesome neighbours, nor is it matched by the power of the state to keep local boundaries in order.

At the end of the day, as we still beat around the bush, we need to ask whether conservationists who often seek to avoid a confrontation with developers, which they cannot hope to win, can win the hearts and minds of landowners whose own achievement of conservation is not exactly what they want. The Occidental donors who fund conservation projects are not in the business of funding the roads, schools or other public goods that would satisfy the most immediate and explicit desires of the rural population. And that is partly why the mediators in the conversation between Western conservationists and Melanesian landowners are backed into a corner where indigenous knowledge gains its peculiar significance. But the celebration of indigenous knowledge by community development strategies that are premised on the existence of a traditional conservation ethic, or the partial exploitation of such knowledge by business development strategies which seek to divert rural villagers from their current economic practices, cannot readily address the basic facts of rural poverty and social inequality.

For this reason, we also need to ask whether conservationists could achieve more of what they want by seeking to modify the behaviour of local communities than they would achieve by either doing nothing at all, leaving landowners alone to suffer in silence, or by using their own scarce resources to change the actions of other characters in the policy process (Dove 1996; Brandon 1997). I

have previously argued that the Occidental donors could carry on their fight with the Oriental loggers by the simple expedient of persuading the national government to cease the dispensation of timber permits in areas which are known to be rich in biodiversity, without necessarily having to fund conservation projects in those areas (Filer and Sekhran 1998: 257–258). The World Bank has actually used its own financial leverage to wrap increasing volumes of red tape around a forest industry which it now regards as a threat to the values of good governance (Filer, Dubash and Kalit 2000), and in this respect, it has helped to create the physical space in which conservation projects are freed from the necessity of competing directly with the logging companies. But if the conservationists have bought more time in which to carry on their dialogue with local landowners, they may still find, at the end of the day, that money talks louder than birds.

References

- Allen, B.J. 1986. Environmental ethics and village agriculture. *Yagl-Ambu* 13 (3): 1–14.
- Ballard, C. 1997. 'It's the land, stupid!' The moral economy of resource ownership in Papua New Guinea. In *The Governance of Common Property in the Pacific Region*. (ed) P. Larmour. Pacific Policy Paper 19. 47–65. Canberra: Australian National University, National Centre for Development Studies.
- BCN (Biodiversity Conservation Network). 1997. *Getting Down to Business: Annual report 1997*. Washington, DC: BCN.
- 1999. *Evaluating Linkages Between Business, the Environment and Local Communities: Final Stories from the Field*. Washington, DC: BCN.
- Brandon, K. 1997. Policy and practical considerations in land-use strategies for biodiversity conservation. In *Last Stand: Protected areas and the defense of tropical biodiversity*. (eds) R. Kramer, C. van Schaik and J. Johnson. New York: Oxford University Press. 90–113.
- Brosius, J.P. 1999. Analyses and interventions: Anthropological engagements with environmentalism. *Current Anthropology* 40: 277–309.
- Bulmer, R. 1982. Traditional conservation practices in Papua New Guinea. In *Traditional Conservation in Papua New Guinea: Implications for today*. (eds) L. Morauta, J. Pernetta and W. Heaney. Monograph 16. 39–77. Boroko: Institute of Applied Social and Economic Research.
- CRI (Christensen Research Institute). 1996a. Report of the First Two Community Development Patrols in the Bismarck-Ramu ICAD Area of Interest. Unpublished report to the PNG Conservation Resource Centre.
- 1996b. Report of the Second Two Community Development Patrols in the Bismarck-Ramu ICAD Area of Interest. Unpublished report to the PNG Conservation Resource Centre.
- 1996c. Report of the Third Round of Community Development Patrols in the Bismarck-Ramu ICAD Area of Interest. Unpublished report to the PNG Conservation Resource Centre.
- 1997a. Report of the Fourth Round of Community Development Patrols in the Bismarck-Ramu ICAD Area of Interest. Unpublished report to the PNG Conservation Resource Centre.

- 1997b. Report of the Seventh Round of Community Development Patrols in the Bismarck-Ramu ICAD Area of Interest. Unpublished report to the PNG Conservation Resource Centre.
- Croll, E. and D. Parkin. 1992. Cultural Understandings of the Environment. In *Bush Base, Forest Farm: Culture, environment and development*. (eds) E. Croll and D. Parkin. 11–36. London: Routledge.
- Diamond, J. 1997. *Guns, Germs and Steel: A short history of everybody for the last 13,000 years*. London: Jonathan Cape.
- Dove, M.R. 1996. Center, periphery, and biodiversity: A paradox of governance and a developmental challenge. In *Valuing Local Knowledge: Indigenous people and intellectual property rights*. (eds) S.B. Brush and D.F. Stabinsky. Washington, DC: Island Press. 41–67.
- Dwyer, P.D. 1982. Wildlife conservation and tradition in the highlands of Papua New Guinea. In *Traditional Conservation in Papua New Guinea: Implications for today*. (eds) L. Morauta, J. Pernetta and W. Heaney. Monograph 16. 173–189. Boroko: Institute of Applied Social and Economic Research.
- 1997. Modern conservation and indigenous peoples: In search of wisdom. *Pacific Conservation Biology* 1: 91–97.
- Ellen, R.F. 1993. Rhetoric, practice and incentive in the face of the changing times: A case study of Nuaulu attitudes to conservation and deforestation. In *Environmentalism: The view from anthropology* (ed.) K. Milton. ASA Monograph 32. 126–143. London: Routledge.
- Ellis, J.-A. 1997 [actually published 1999]. *Race for the Rainforest 2: Applying lessons learnt to the Bismarck-Ramu integrated conservation and development process in Papua New Guinea*. Port Moresby: PNG Biodiversity Conservation and Resource Management Programme.
- Filer, C. 1991a. The eyes of the world are on Lak. *Times of PNG*. 3 and 10 January.
- 1991b. Two shots in the dark: The first year of the task force on environmental planning in priority forest areas. *Research in Melanesia* 15(1): 1–48.
- 1994. The nature of the human threat to Papua New Guinea's biodiversity endowment. In *Papua New Guinea Country Study on Biological Diversity*. (eds.) N. Sekhran and S. Miller. Waigani: PNG Department of Environment and Conservation. Nairobi: Africa Centre for Resources and Environment. 187–199.
- 1997. Compensation, rent and power in Papua New Guinea. In *Compensation for Resource Development in Papua New Guinea*. (ed.) S. Toft. Monograph 6. Boroko: Law Reform Commission. Canberra: Australian National University, National Centre for Development Studies (Pacific Policy Paper 24). 156–189.
- Filer, C. with N. Sekhran. 1998. *Loggers, Donors and Resource Owners*. London: International Institute for Environment and Development in association with the National Research Institute (Policy That Works for Forests and People, Papua New Guinea Country Study).
- Filer, C. with N.K. Dubash and K. Kalit. 2000. *The Thin Green Line: World Bank leverage and forest policy reform in Papua New Guinea*. Monograph 37. Boroko: National Research Institute. Canberra: Australian National University, Research School of Pacific and Asian Studies, Resource Management in Asia-Pacific Project.
- Filer, C., M. Hedemark, A. Kituai, S. Majnep and C. Unkau. 1995. *Ramu Conservation Area: April 1995 site visit*. Unpublished report to the PNG Conservation Resource Centre.

- Filer, C. and W. Iamo. 1989. *Base-Line Planning Study for the Lakekamu Gold Project*. Unpublished report to the PNG Department of Minerals and Energy.
- Grant, N. 1996. *Community Entry for ICAD Projects – The participatory way*. Port Moresby: PNG Biodiversity Conservation and Resource Management Programme.
- Grillo, R.D. and R.L. Stirrat. (eds) 1997. *Discourses of Development: Anthropological perspectives*. Oxford and New York: Berg.
- Kirsch, S. 1997. Regional dynamics and conservation in Papua New Guinea: The Lakekamu River Basin Project. *Contemporary Pacific* 9 (1): 97–121.
- Kocher Schmid, C. and S. Klappa. 1999. Profile of a leader, or the world according to Yulu Nuo. In *Expecting the Day of Wrath: Versions of the millennium in Papua New Guinea*. (ed.) C. Kocher Schmid. Monograph 36. Boroko: National Research Institute. 89–109.
- Lalley, B. 1998. Is conservation the answer? In *The Motupore Conference: ICAD practitioners views from the field*. (eds) S.M. Saulei and J.-A. Ellis. Waigani: PNG Biodiversity Conservation and Resource Management Programme. 16–19.
- Margoluis, R. and N. Salafsky. 1998. *Measures of Success: Designing, managing and monitoring conservation and development projects*. Washington, DC: Island Press.
- McCallum, R. and N. Sekhran. 1997. *Race for the rainforest: Evaluating lessons from an integrated conservation and development experiment in New Ireland, Papua New Guinea*. Port Moresby: PNG Biodiversity Conservation and Resource Management Programme.
- Nelson, H. 1976. *Black, White and Gold: Goldmining in Papua New Guinea, 1878–1930*. Canberra: Australian National University Press.
- Orsak, L. 1997. *Twenty Arguments for NOT Using Natural Resources Sustainably and the Weaknesses with these Arguments: Villager perspectives in Papua New Guinea*. Madang: Christensen Research Institute.
- Saulei, S.M. and J.-A. Ellis (eds). 1998. *The Motupore Conference: ICAD practitioners views from the field*. Waigani: PNG Biodiversity Conservation and Resource Management Programme.
- Van Helden, F. 1998a. Between cash and conviction: The social context of the Bismarck-Ramu Integrated Conservation and Development Project. Monograph 33. Boroko: National Research Institute.
- 1998b. *The Bismarck-Ramu Social Feasibility Study: Overview and recommendations*. Unpublished report to the PNG Biodiversity Conservation and Resource Management Programme.
- n.d. The Community entry approach of the Bismarck-Ramu Integrated Conservation and Development Project. To be published in *Custom, Conservation and Development in Papua New Guinea*. (ed.) C. Filer. *Forthcoming*.
- West, C.P. 2000. *The Practices, Ideologies, and Consequences of Conservation and Development in Papua New Guinea*. New Brunswick, NJ: Rutgers University (PhD thesis).
- World Bank. 1991. Operational directive 4.20: *Indigenous Peoples*. Washington, DC: World Bank.
- Young, M. 1991. Logging or conservation on Woodlark (Muyuw) Island. *Research in Melanesia* 15: 49–65.
- Zerner, C. 1996. Telling stories about biodiversity. In *Valuing Local Knowledge: Indigenous people and intellectual property rights*. (eds) S.B. Brush and D.F. Stabinsky. Washington, DC: Island Press. 68–101.